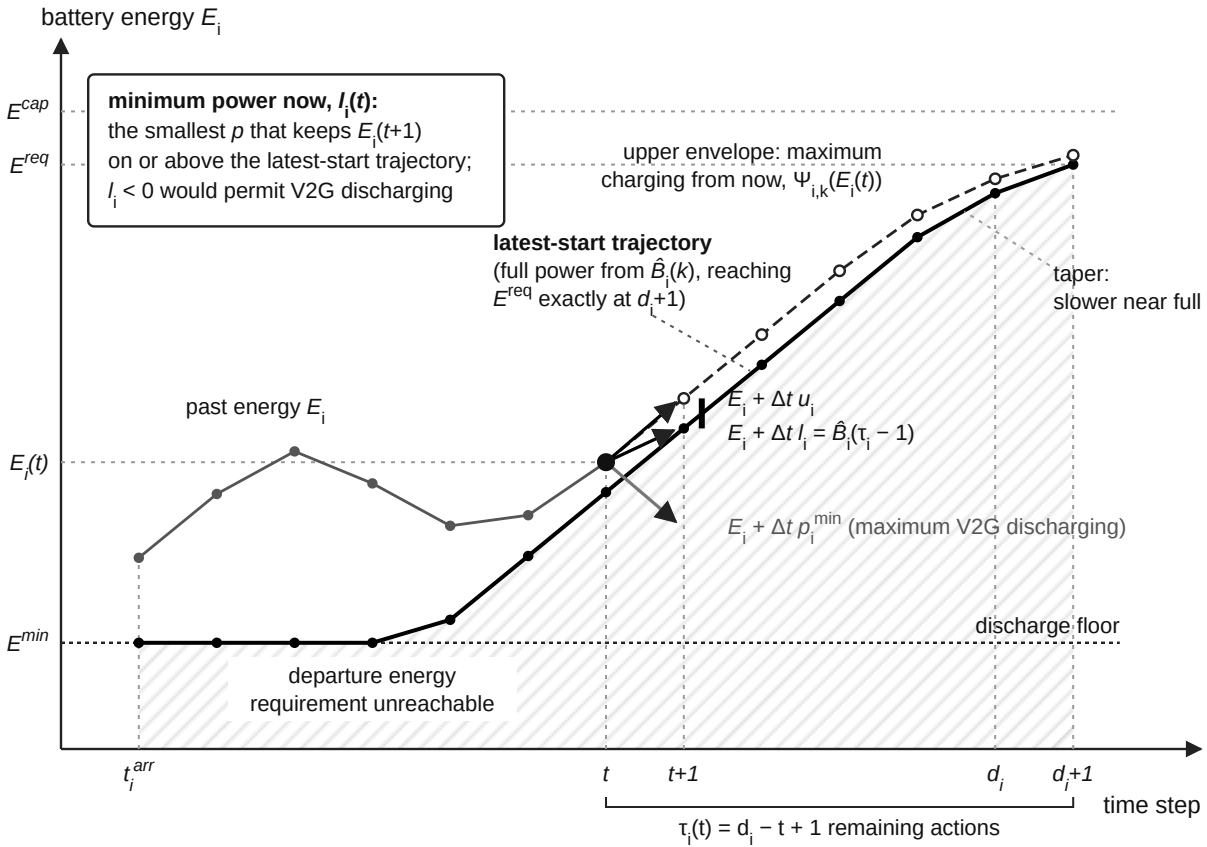

battery energy E_i
minimum power now, $l_i(t)$:
the smallest p that keeps $E_i(t+1)$
on or above the latest-start trajectory;
$l_i < 0$ would permit V2G discharging
E^{cap}
E^{req}
upper envelope: maximum
charging from now, $\Psi_{i,k}(E_i(t))$
latest-start trajectory
(full power from $\hat{B}_i(k)$, reaching
E^{req} exactly at d_i+1)
taper:
slower near full
past energy E_i
$E_i + \Delta t\, u_i$
$E_i + \Delta t\, l_i = \hat{B}_i(\tau_i - 1)$
$E_i(t)$
$E_i + \Delta t\, p_i^{min}$ (maximum V2G discharging)
E^{min}
discharge floor
departure energy
requirement unreachable
t_i^{arr}
t
$t+1$
d_i
d_i+1
time step
$\tau_i(t) = d_i - t + 1$ remaining actions